MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 12ARP19

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Dual beam electrostatic focus and deflection, post-accelerator

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE RATINGS:

Parameter: Unit	Ef V	Ec1 Vdc	Eb1 Vdc	Eb2 Vdc (Note 1)	Eb3 Vdc	ed v	Rg Meg	Zd Meg (Note 2)	Ehk Vdc
Maximum:	6.9	0	4,400	11,000	13,200	2,500	1.5	5.0	+180
Minimum:	5.7	-300	---	5,000	5,000	---	---	---	-180
TEST CONDITIONS:	6.3	Adj	Focus	7,500	7,700	---	---	---	---

GENERAL:

Qualification - Required

Ⓐ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection	3					
5216	Cathode illumination	-		---	---	---	---
5248	Deflection-factor uniformity	-		---	---	---	---
1331	Direct-interelectrode capacitance	-	Cathode to all	Ck	---	6.5	pF
			Grid 1 to all	Cg1	---	6.5	pF
			D1 to D2	C1D2	---	4.0	pF
			D3 to D4	C3D4	---	3.0	pF
			D1 to all	CD1	---	9.0	pF
			D2 to all	CD2	---	9.0	pF
			D3 to all	CD3	---	9.5	pF
			D4 to all	CD4	---	9.5	pF
1141	Pressure (implosion)	-		---	---	---	---
5250	Interaction factor	4		---	$14X10^{-5}$		Vdc/in.
5111	Vibration	-		Width	---	2	mm
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas "cross"	5	Ib3 = 2 μAdc	---	---	---	- -
5223	Modulation	-		ΔEc1	---	16	Vdc
5231	Spot position (electro-static deflection)	-		Displ	---	16	mm
5231	Spot displacement (leakage)	-		Displ	---	10	mm
5241	Grid-cutoff voltage	-		Ec1	-150	-225	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No. 1 leakage current	-		---	---	---	---
5251	Anode No. 2 leakage current	-		---	---	---	---
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
	Quality conformance inspection, part 2	3					
1301	Heater current	-		If	540	660	mA
5201	Electrode current (anode No. 1)	-	Ib3 = 2 μAdc	Ib1	-10	+5	μAdc
5101	Base alignment (electrostatic types)	-	+1D2, pin No. 4	---	---	---	---
5101	Side terminal alignment (electrostatic types)	-	+1D2	---	---	---	---
5101	Side terminal and base alignment	-	Pin No. 4	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS MINIMUM	LIMITS MAXIMUM	UNIT
	Quality conformance inspection, part 2 -Continued	3					
5101	Neck and base alinement (electrostatic types)	-		---	---	---	---
5216	Stray light emission (conventional types)	-	Eb1 = 4,000 V; Eb2 = 10,000 V; Eb3 = 13,200 V	---	---	---	---
5246	Focusing voltage	-		Eb1	1,750	2,500	Vdc
---	Deflection defocusing	7		---	---	---	---
---	Deflection defocusing	8		---	---	---	---
---	Tracking accuracy	9		---	---	---	---
5251	Heater-cathode leakage current	-		---	---	---	---
(A) 5101	Angle between traces	-		---	89	91	Degrees
(A) 5101	Angle between corresponding traces	-		---	---	---	---
(A) 5221	Screen (P19)	6		cBf D0.1 D1.0 D1.0-D10.0	575 410 335 75	--- --- --- 120	cB cB cB cB
(A) 5226	Line width A	-	Ib3 = 2 uAdc	Width	---	0.40	mm
(A) 5226	Line width B	-	Ib3 = 2 uAdc	Width	---	0.50	mm
(A) 5246	Focusing voltage at cutoff	-		Eb1	1,750	2,500	Vdc
(A) 5248	Deflection factor	-	1D2	DF	130	177	Vdc/in.
(A) 5248	Deflection factor	-	3D4	DF	130	177	Vdc/in.
	Quality conformance inspection, part 3						
---	Life test	-	Group C; Ib = 2 uA/gun; t = 500 hours (min)	---	---	---	---
---	Life-test end points:						
5223	Modulation	-	Ib3 = 2 uAdc	ΔEc1	---	10	Vdc
5226	Line width A	-	Ib3 = 2 uAdc	Width	---	0.46	mm
5226	Line width B	-	Ib3 = 2 uAdc	Width	---	0.58	mm

NOTES:

1. Accelerator power input (average) should be limited to 6 watts.
2. It is recommended that the deflection electrode circuit resistances be approximately equal. Higher resistance values up to 5.0 megohms may be used for low-beam current operation.
3. All tests except capacitance, vibration, tracking, cathode illumination, stray emission, and interaction factor shall be made on each unit separately.
4. The deflection per unit voltage of one beam at any position of the beam within its useful scan when a balanced voltage of at least ±300 volts dc is applied to the 3D4 terminals of the other gun shall be not greater than the value specified.

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5. This test to be performed at the conclusion of the holding period.

6. The phosphorescent light intensity observed t seconds after completion of the application of the nth raster to the tube is Dt(n). cBf shall be the fluorescent light output under continuous raster excitation. The screen characteristics shall be measured under the following conditions:

   (a) Raster size (focused beam): 7.1 cm by 7.1 cm giving a total area of 50 cm^2. The tube shall be defocused to a spot approximately 0.25 cm in diameter.
   (b) Beam current: 2 μA at tube operating voltage of Eb3 = 10,000V.
   (c) Duration of raster: 1/60 second. The raster is formed by means of linear scanning frequency of 12 kHz and the vertical scanning frequency of 60 Hz. The raster is repeated at 1 second intervals.
   (d) Distance of raster from axis of the calibrated 931VA multiplier tube: 30 cm.
   (e) White light: Sufficient to de-excite the phosphor before build up measurements so that the results are within 3 cB of those that would have been obtained after total de-excitation.
   (f) After the tenth raster, excitation shall be removed. The light output 0.1 second (D0.1), 1 second (D1.0), and 10 seconds (D10.0) after excitation is removed shall satisfy the specified limits. Before performing the above measurements, illuminate the screen with a 250 watt projection lamp for approximately 1 minute, then wait 30 seconds after lamp is turned off. A description of the primary standard replica lamps for the various screen types will be found in the Material Laboratory, New York Naval Shipyard, Final Report 5032-B-15.40, dated 22 July 1959. Centibels may be converted to Milli-foot lamberts by the use of the following equation.

$$MftL = 1.77 \times 10^{-3} \times 10^{\frac{cB}{100}}$$

7. Variation in spot size anywhere within a circle 11 inches (279.40 mm) in diameter will not exceed 1.5:1 under compromise focus conditions.

8. When a low intensity dc spot is focused at the center of the tube face and deflected to the edges of scan without readjustment of focus voltage, the major dimension of the spot shall not exceed 0.070 inch (1.78 mm) anywhere within a circle 11 inches (279.40 mm) in diameter.

9. The measurement of tracking error shall be made as follows:

   (a) The focused spots of both guns shall be removed by means of a common dc voltage along the 1D2 axis until the spot of gun A is at +3 inches (76.20 mm) from the center of the tube. Then by means of a small correction voltage the spot of gun B shall be moved to the same 1D2 position. The spots then shall be moved by means of the common dc voltage in the opposite direction until the spot of the less sensitive gun reaches -3 inches (76.20 mm) by means of a voltage divider. The procedure is repeated until 1D2 deflection factor compensation is complete. The entire procedure is then duplicated along the 3D4 axis with a distance from the center of ± 3 inches (76.20 mm). Then by means of the two common dc voltages the spots shall be tracked over the area within a circle 11 inches (279.40 mm) in diameter. If desired a small correction voltage may be applied to the plates of one gun to allow optimum spot positioning for minimum over-all tracking error.
   (b) The separation of spots over the tracking area shall not exceed 0.15 inch (3.81 mm).

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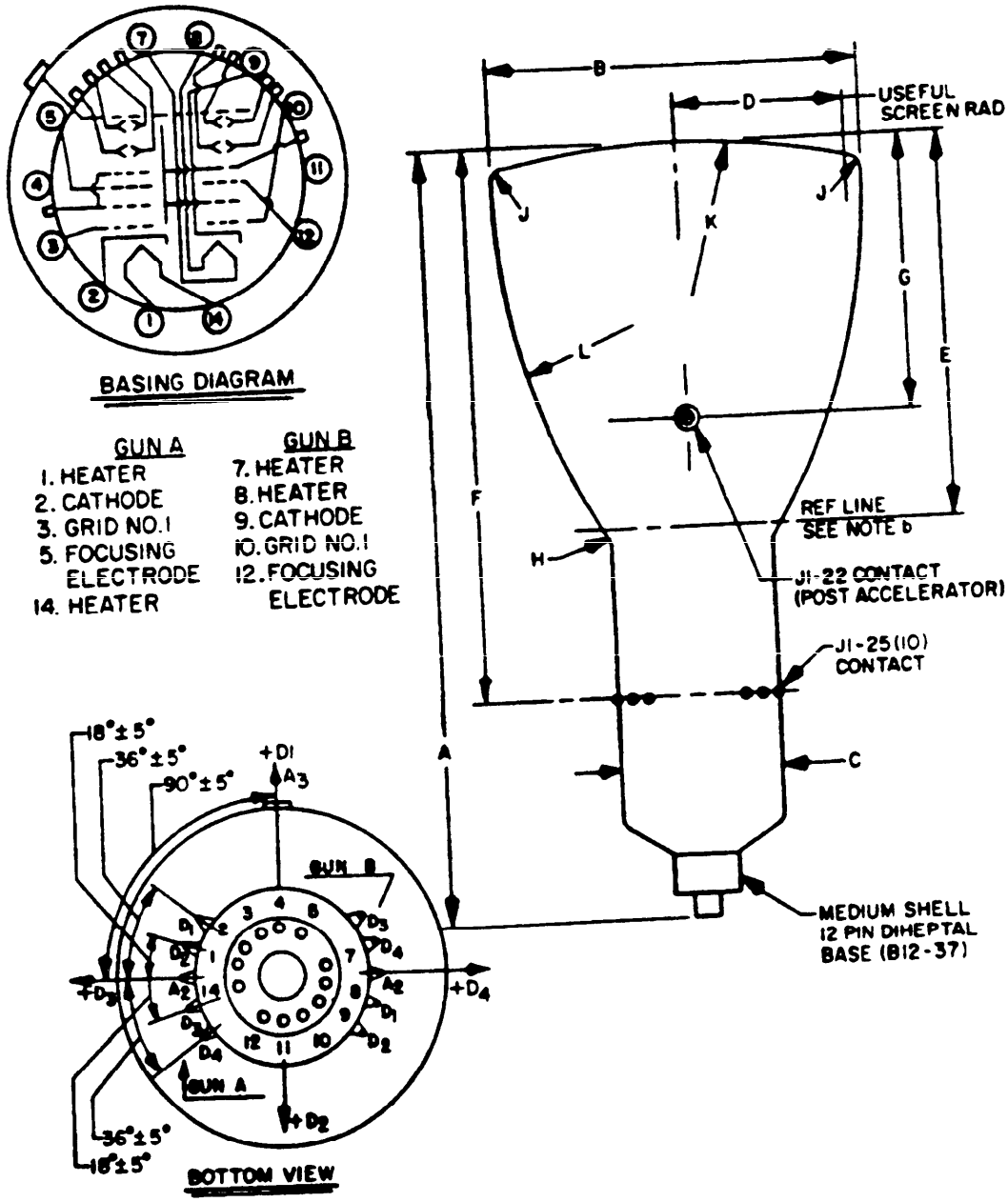


FIGURE 1. Outline drawing of electron tube type 12ARP19.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
	Qualification inspection	
	Base: B12-37	
	Contacts: J1-22 (1) J1-25 (10)	
	Quality conformance inspection, part 2	
A	24.50 (622.30)	25.25 (641.35)
B	12.38 (314.45)	12.50 (317.50)
C	5.15 (130.81)	5.34 (135.64)
Ⓐ D	5.25 (133.35) R	---
E	12.25 (311.15)	12.54 (318.52)
F	17.50 (444.50)	18.00 (457.20)
G	8.50 (215.90)	9.00 (228.60)
H	---	1.00 (25.40) R
	Reference dimensions (see note c)	
J	.44 (11.18) R	
K	40 (1016.00) R	
L	24 (609.60) R	

NOTES:
a. Base pin No. 4 alines with J1-22 contact.
b. Reference line is that point where a 5.750 ±.003 (146.05 ±.08 mm) diameter ring gage will stop.
c. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 12ARP19 - Continued.

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